جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

FACULTY OF COMPUTING
& INFORMATION SCIENCES

C-SW311: Software Design and Development
Spring 2026

Submission and Discussion guidelines

Guidelines for project deliverables

1. For all diagrams, be **creative and rationale** on your assumptions about the information required, and try to include everything that is *important* for your model to be explanatory.
2. For EACH deliverable, each group must submit a well-written and organized **Technical Report document** containing ALL steps, tables and diagrams and also describing your solutions and rationales to the assignment (based on the deliverable requirements, preferably zipped or otherwise compressed). **Any assumptions you made during your work must be explicitly mentioned either in the Technical Report**.
3. Submit your **Technical Report document** in PDF format and No Handwriting will be accepted.
4. All submissions will be on Canvas and a submission link will be created for each deliverable.
5. **Plagiarism will be treated strictly.**
6. **NO LATE Submission will be accepted, and NO EXCEUSES**
7. **Any late submission will take ZERO.**
8. Please bear in mind that submission at the last minute might cause a network problem, and that would not be taken as an excuse. Therefore, you need to submit as early as possible on the submission day.
9. Discussions will be scheduled after the submission. Eng. Tawfik and Eng. Mohammed will return the submitted reports to each group with feedback highlighted inline (in the form of embedded comments).

جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

FACULTY OF COMPUTING
& INFORMATION SCIENCES

Guidelines for Deliverable#3: Behavioural Design Patterns

Deadline of submission: Thursday 16th of April, 2026 at 11:59 pm.

1. **From your project context, identify cases or problems that require the use of the following patterns:**

    a. **Template Method Pattern:**
    Identify **two cases/problems** where the Template Method pattern can be effectively applied.

    b. **Mediator Pattern:**
    Identify **two cases/problems** where the Mediator pattern should be used to manage interactions between components.

    c. **Strategy Pattern:**
    Identify **two cases/problems** where the Strategy pattern would be beneficial for defining interchangeable behaviors.

    d. **Observer + Singleton Patterns:**
    Identify **one case/problem** where a combination of Observer and Singleton patterns can be applied.

    e. **Iterator Pattern:**
    Identify **one case/problem** where the Iterator pattern would be useful for traversing a collection.

2. **For Each Case/Problem Identified Above**

For every design pattern case identified in Section 1:

a. **UML Design:** Develop a **UML class diagram** that conceptually represents your proposed solution.
b. **Implementation:** Implement the solution in **Java within your project**, including the Client/Usage of the pattern

Note: You are **not required to include the code in the report**. However, the implementation must be completed as part of your working project.

Deliverable Type

Students are required to submit the following:

a) **Report**

- A structured document explaining the identified cases/problems for each required creational design pattern.
- UML class diagrams for each case.
- Clear justification for why each pattern was selected and how it solves the identified problem.

b) **NetBeans Project**

- The complete and functional Java project.
- All implemented design patterns must be properly integrated.
- The project must compile and run without errors.
- The development environment should reflect the required DevOps tools (e.g., Maven configuration, Git repository usage, etc.).

Please ensure that both components are submitted before the deadline.